

KINATRAX Technical Assessment

Context:

We've collected motion capture and ball-tracking data from a professional pitcher recorded in 2023 and again in 2025. The dataset includes both **biomechanical metrics** (e.g., hip rotation, shoulder rotation, elbow extension) and **pitch metrics** (e.g., release velocity, spin rate, movement, release position).

Your task is to create a **React dashboard** that explores and visualizes trends or insights you find interesting about this pitcher's data in both years. Our tools are developed with the coach and player in mind, so think about how they might best understand the data. There are no wrong ways of going about it! We are curious to see what you find interesting.

Data

You'll receive two JSON files containing both pitch-level and biomechanics related metrics. A broader dictionary of the two files can be found in the Kinatrax Data Dictionary google sheet.

- **pitch_biomech_data.json**: This file contains pitch level discrete data describing both play state (i.e. **inning**, **pitch number**, **batter side**, **pitcher side**), ball metrics (i.e. **pitch speed**, **horizontal break**, **vertical break**, **spin rate**, etc), and biomech data (i.e. **Release Hand Position**, **Elbow Flexion**, etc.)
 - **time_series_metrics.json**: For every PitchUID in **pitch_biomech_data** are 10 different time series metrics at the frame-level such as **Pitching_Shoulder_Rotation**, **Pitching_Shoulder_Rotation**, **Front_Foot_CG_Pos**, **Center_Mass_Forward**
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Your Task

Build a **React app** that:

1. Loads and visualizes the dataset interactively.
2. Presents the data in a way in which the user can gain insight
3. Lets a user explore the data — through charts, filters, toggles, or whatever interface you think makes sense.

Suggested Directions (choose one or mix several)

You have creative freedom — but here are a couple of prompts to get started:

- Compare **biomech changes** in 2023 vs 2025 data (e.g., release point, shoulder rotation, etc).
 - Examine **pitch outcome trends** (velocity/spin rate changes, horizontal/vertical break profile changes) over time.
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Requirements

- Use **React/Vite** for development
 - Include at least two distinct types of visualizations (e.g. time series + scatter, heatmap + bar, table viewing).
 - Be prepared to walk-thru your findings in a subsequent interview if warranted
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Evaluation Criteria

Category	What We’re Looking For
Technical implementation	React component structure, data handling, reusability
Visualization quality	Clarity, labeling, interactivity
Insight & reasoning	Ability to derive meaningful observations from the data
Design & UX	Layout, responsiveness, usability
Polish / creativity	Extra features (animations, filters, summaries)

Deliverables

- A small README or writeup summarizing your design choices and main takeaways about any changes the data may show.
- A GitHub repo or zip of your React app